



# Units That Cancel

## **Grades 9-12 • SAT / ACT mathematics foundations**

Problem-Solving and Data Analysis • Use dimensional reasoning to convert a rate.

Original focused skill practice for learners preparing for college-entrance mathematics. Includes models, 16 practice tasks, six explain-and-check tasks, a four-item exit ticket, and worked solutions. Not a full practice exam.

**PRINT • TEACH • REVISIT**



# Plan a focused study session

## Learning objective

Use dimensional reasoning to convert a rate.

## Readiness

Use after learners have studied the relevant algebra, geometry, or data concepts. Grade tags describe a typical preparation audience; readiness depends on prior coursework.

## Suggested use

Session 1: 10 minutes on worked examples and 20–30 minutes on practice. Session 2: review errors, then use explain-and-check tasks. Finish with the independent exit ticket. Pause and reteach a prerequisite when the model is not yet understood.

## What this resource covers

The named mathematics skill is useful within SAT and ACT preparation. The packet does not cover every exam domain, reproduce an official section, provide standardized timing, or predict a score.

## Materials

Pencil, paper, and a calculator for checking when useful. Practice both symbolic reasoning and calculator verification. Follow each exam publisher's current policies for an actual test.

## Student directions

Read the entire question, define quantities, show a method, and check the result. For a written explanation, cite the calculation or feature that supports your conclusion.



## Two worked examples

**Model 1. A cart moves at 5 meters per second for 6 minutes at a constant speed. How many kilometers does it travel? Show units canceling.**

$$5 \text{ m/s} \times 6 \text{ min} \times 60 \text{ s/min} \times 1 \text{ km}/1000 \text{ m} = 9/5 \text{ km}.$$

**Model 2. A cart moves at 3 meters per second for 11 minutes at a constant speed. How many kilometers does it travel? Show units canceling.**

$$3 \text{ m/s} \times 11 \text{ min} \times 60 \text{ s/min} \times 1 \text{ km}/1000 \text{ m} = 99/50 \text{ km}.$$

Pause before looking at a solution: which relationship, definition, or constraint determines the first step?

# Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

1. A cart moves at 12 meters per second for 10 minutes at a constant speed. How many kilometers does it travel? Show units canceling.

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2. A cart moves at 8 meters per second for 9 minutes at a constant speed. How many kilometers does it travel? Show units canceling.

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3. A cart moves at 12 meters per second for 9 minutes at a constant speed. How many kilometers does it travel? Show units canceling.

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# Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

4. A cart moves at 5 meters per second for 10 minutes at a constant speed. How many kilometers does it travel? Show units canceling.

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5. A cart moves at 5 meters per second for 8 minutes at a constant speed. How many kilometers does it travel? Show units canceling.

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6. A cart moves at 12 meters per second for 7 minutes at a constant speed. How many kilometers does it travel? Show units canceling.

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## Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

7. A cart moves at 8 meters per second for 6 minutes at a constant speed. How many kilometers does it travel? Show units canceling.

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8. A cart moves at 8 meters per second for 11 minutes at a constant speed. How many kilometers does it travel? Show units canceling.

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9. A cart moves at 5 meters per second for 7 minutes at a constant speed. How many kilometers does it travel? Show units canceling.

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## Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

10. A cart moves at 8 meters per second for 7 minutes at a constant speed. How many kilometers does it travel? Show units canceling.

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11. A cart moves at 5 meters per second for 9 minutes at a constant speed. How many kilometers does it travel? Show units canceling.

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12. A cart moves at 12 meters per second for 7 minutes at a constant speed. How many kilometers does it travel? Show units canceling.

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## Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

13. A cart moves at 12 meters per second for 13 minutes at a constant speed. How many kilometers does it travel? Show units canceling.

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14. A cart moves at 12 meters per second for 6 minutes at a constant speed. How many kilometers does it travel? Show units canceling.

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15. A cart moves at 5 meters per second for 7 minutes at a constant speed. How many kilometers does it travel? Show units canceling.

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## Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

16. A cart moves at 3 meters per second for 10 minutes at a constant speed. How many kilometers does it travel? Show units canceling.

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# Explain and check • Transfer practice

Solve the task, then explain a check or limitation. Compare your reasoning with the worked key only after making your own attempt.

1. A cart moves at 8 meters per second for 9 minutes at a constant speed. How many kilometers does it travel? Show units canceling.

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2. A cart moves at 3 meters per second for 6 minutes at a constant speed. How many kilometers does it travel? Show units canceling.

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# Explain and check • Transfer practice

Solve the task, then explain a check or limitation. Compare your reasoning with the worked key only after making your own attempt.

3. A cart moves at 5 meters per second for 12 minutes at a constant speed. How many kilometers does it travel? Show units canceling.

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4. A cart moves at 3 meters per second for 11 minutes at a constant speed. How many kilometers does it travel? Show units canceling.

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# Explain and check • Transfer practice

Solve the task, then explain a check or limitation. Compare your reasoning with the worked key only after making your own attempt.

5. A cart moves at 8 meters per second for 7 minutes at a constant speed. How many kilometers does it travel? Show units canceling.

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6. A cart moves at 5 meters per second for 12 minutes at a constant speed. How many kilometers does it travel? Show units canceling.

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# Exit ticket • Independent check

Complete independently. Show reasoning and any relevant restriction. This is a classroom learning check, not a standardized test score.

1. A cart moves at 8 meters per second for 8 minutes at a constant speed. How many kilometers does it travel? Show units canceling.

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2. A cart moves at 5 meters per second for 10 minutes at a constant speed. How many kilometers does it travel? Show units canceling.

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# Exit ticket • Independent check

Complete independently. Show reasoning and any relevant restriction. This is a classroom learning check, not a standardized test score.

3. A cart moves at 8 meters per second for 9 minutes at a constant speed. How many kilometers does it travel? Show units canceling.

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4. A cart moves at 3 meters per second for 9 minutes at a constant speed. How many kilometers does it travel? Show units canceling.

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# Practice worked solutions

## 1. 36/5 km

$$12 \text{ m/s} \times 10 \text{ min} \times 60 \text{ s/min} \times 1 \text{ km/1000 m} = 36/5 \text{ km.}$$

## 2. 108/25 km

$$8 \text{ m/s} \times 9 \text{ min} \times 60 \text{ s/min} \times 1 \text{ km/1000 m} = 108/25 \text{ km.}$$

## 3. 162/25 km

$$12 \text{ m/s} \times 9 \text{ min} \times 60 \text{ s/min} \times 1 \text{ km/1000 m} = 162/25 \text{ km.}$$

## 4. 3 km

$$5 \text{ m/s} \times 10 \text{ min} \times 60 \text{ s/min} \times 1 \text{ km/1000 m} = 3 \text{ km.}$$

# Practice worked solutions

## 5. 12/5 km

$$5 \text{ m/s} \times 8 \text{ min} \times 60 \text{ s/min} \times 1 \text{ km/1000 m} = 12/5 \text{ km.}$$

## 6. 126/25 km

$$12 \text{ m/s} \times 7 \text{ min} \times 60 \text{ s/min} \times 1 \text{ km/1000 m} = 126/25 \text{ km.}$$

## 7. 72/25 km

$$8 \text{ m/s} \times 6 \text{ min} \times 60 \text{ s/min} \times 1 \text{ km/1000 m} = 72/25 \text{ km.}$$

## 8. 132/25 km

$$8 \text{ m/s} \times 11 \text{ min} \times 60 \text{ s/min} \times 1 \text{ km/1000 m} = 132/25 \text{ km.}$$



# Practice worked solutions

## 9. 21/10 km

$$5 \text{ m/s} \times 7 \text{ min} \times 60 \text{ s/min} \times 1 \text{ km/1000 m} = 21/10 \text{ km.}$$

## 10. 84/25 km

$$8 \text{ m/s} \times 7 \text{ min} \times 60 \text{ s/min} \times 1 \text{ km/1000 m} = 84/25 \text{ km.}$$

## 11. 27/10 km

$$5 \text{ m/s} \times 9 \text{ min} \times 60 \text{ s/min} \times 1 \text{ km/1000 m} = 27/10 \text{ km.}$$

## 12. 126/25 km

$$12 \text{ m/s} \times 7 \text{ min} \times 60 \text{ s/min} \times 1 \text{ km/1000 m} = 126/25 \text{ km.}$$

# Practice worked solutions

## 13. 234/25 km

$$12 \text{ m/s} \times 13 \text{ min} \times 60 \text{ s/min} \times 1 \text{ km/1000 m} = 234/25 \text{ km.}$$

## 14. 108/25 km

$$12 \text{ m/s} \times 6 \text{ min} \times 60 \text{ s/min} \times 1 \text{ km/1000 m} = 108/25 \text{ km.}$$

## 15. 21/10 km

$$5 \text{ m/s} \times 7 \text{ min} \times 60 \text{ s/min} \times 1 \text{ km/1000 m} = 21/10 \text{ km.}$$

## 16. 9/5 km

$$3 \text{ m/s} \times 10 \text{ min} \times 60 \text{ s/min} \times 1 \text{ km/1000 m} = 9/5 \text{ km.}$$

# Transfer worked solutions

## 1. 108/25 km

$$8 \text{ m/s} \times 9 \text{ min} \times 60 \text{ s/min} \times 1 \text{ km/1000 m} = 108/25 \text{ km.}$$

## 2. 27/25 km

$$3 \text{ m/s} \times 6 \text{ min} \times 60 \text{ s/min} \times 1 \text{ km/1000 m} = 27/25 \text{ km.}$$

## 3. 18/5 km

$$5 \text{ m/s} \times 12 \text{ min} \times 60 \text{ s/min} \times 1 \text{ km/1000 m} = 18/5 \text{ km.}$$

## 4. 99/50 km

$$3 \text{ m/s} \times 11 \text{ min} \times 60 \text{ s/min} \times 1 \text{ km/1000 m} = 99/50 \text{ km.}$$

# Transfer worked solutions

## 5. 84/25 km

$$8 \text{ m/s} \times 7 \text{ min} \times 60 \text{ s/min} \times 1 \text{ km/1000 m} = 84/25 \text{ km.}$$

## 6. 18/5 km

$$5 \text{ m/s} \times 12 \text{ min} \times 60 \text{ s/min} \times 1 \text{ km/1000 m} = 18/5 \text{ km.}$$

# Exit-ticket worked solutions

## 1. 96/25 km

$$8 \text{ m/s} \times 8 \text{ min} \times 60 \text{ s/min} \times 1 \text{ km/1000 m} = 96/25 \text{ km.}$$

## 2. 3 km

$$5 \text{ m/s} \times 10 \text{ min} \times 60 \text{ s/min} \times 1 \text{ km/1000 m} = 3 \text{ km.}$$

## 3. 108/25 km

$$8 \text{ m/s} \times 9 \text{ min} \times 60 \text{ s/min} \times 1 \text{ km/1000 m} = 108/25 \text{ km.}$$

## 4. 81/50 km

$$3 \text{ m/s} \times 9 \text{ min} \times 60 \text{ s/min} \times 1 \text{ km/1000 m} = 81/50 \text{ km.}$$

# Use errors to choose the next step

## Identify the first mismatch

Compare the student's representation with the quantities or stated condition. Separate reading the situation, choosing a method, calculation, and interpretation.

## Respond precisely

If the model is wrong, use a simpler example before repeating the procedure. If the method is sound and arithmetic is wrong, correct that step and verify the original result.

## Review the exit ticket

For each item: 2 points for an accurate result with coherent reasoning; 1 for a sound method with a minor error or an unexplained correct answer; 0 for unsupported or missing reasoning. Use comments to select a next action. These are local instructional criteria.

## AIG

Compare two methods, construct a counterexample, or create a new problem with a prescribed solution and justify its restrictions.

## 504 / EC

Follow the individual plan. Offer manageable item groups, documented time or format supports, a worked step, or an alternate response method while retaining the mathematical objective.

## ESL

Preview terms and symbols in the questions. Use home-language rehearsal or a bilingual glossary, then ask for a concise explanation such as "The condition means \_\_\_, so I used \_\_\_."

# Sources, preparation, and license

Objective: Use dimensional reasoning to convert a rate. State alignment: not claimed. This is a focused college-entrance preparation resource, not a state course or a claim of course equivalency.

Official exam scope consulted September 15, 2026: College Board, Types of Math Tested, <https://satsuite.collegeboard.org/sat/whats-on-the-test/math/types> ; ACT, Mathematics Test Description, <https://www.act.org/content/act/en/products-and-services/the-act/test-preparation/description-of-math-test.html> .

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